

This adds the elements package required later for building the Web component. Next, type the following:

```
ng generate component MyWebComponent
```

The Angular component is named MyWebComponent. Now, open your favourite typescript editor (e.g., VS Code) and navigate to the angular-web-component directory. You can do it by typing 'code .' in \$pwd. One can download VS Code from <https://code.visualstudio.com/download>. The next command is:

```
npm start
```

Now, open localhost:4200 in your Web browser. In the browser, you can see your first Angular application running. The screen will look something like what's shown in Figure 1.

Developing an Angular component

Now that we have installed Angular and have a 'hello world' sort of application up and running, let's look at how to modify it to something meaningful.

We will do a simple modification to show a REST API call, and render the REST API output in the GUI in a very simple manner. For the REST API part, we will call an open source REST API interface provided by <https://jsonplaceholder.typicode.com/posts>. It will return a tuple consisting of <postID, userID, topic>. We will render it in the GUI with a very simple HTML table, with no fancy grid.

We need to modify the files under the 'src/app/my-web-component' directory as follows.

File: my-web-component.component.html

```
<div style="margin: 10px; text-align: center;">
  <h2> Posts: </h2>
  <table class="table-bordered">
    <tr class="inventory-table-header">
      <th class="text-center">Post ID</th>
      <th class="text-center">User ID</th>
      <th class="text-center">Title</th>
    </tr>
    <tr *ngFor="let post of posts">
      <td>{{post.id}}</td>
      <td>{{post.userId}}</td>
      <td>{{post.title}}</td>
    </tr>
  </table>
</div>
```

File: my-web-component.component.ts

```
import { Component, OnInit } from '@angular/core';
import { HttpClient } from '@angular/common/http';
```

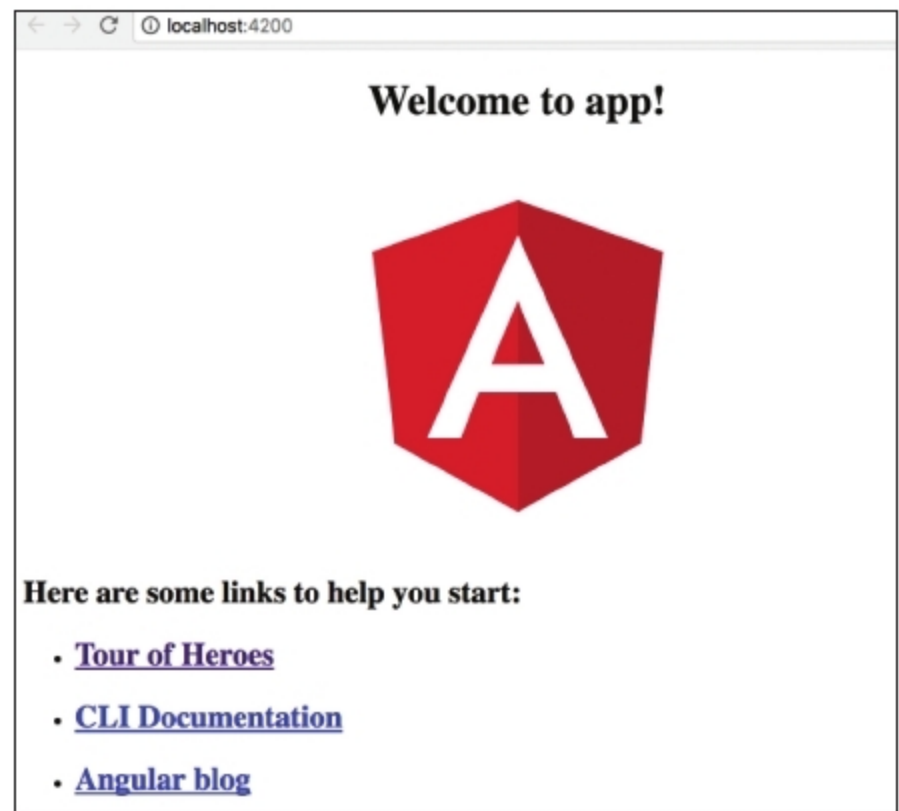


Figure 1: The first Angular app output

```
@Component({
  selector: 'app-my-web-component',
  templateUrl: './my-web-component.component.html',
  styleUrls: ['./my-web-component.component.scss']
})
export class MyWebComponent implements OnInit {

  public posts = [];

  constructor(private httpClient: HttpClient) {
  }

  ngOnInit() {
    this.httpClient.get('https://jsonplaceholder.typicode.com/posts').subscribe((restApiResponse: any) => {
      this.posts = restApiResponse;
    });
  }
}
```

That's enough for the component. There's no need to change the styling file (.scss) and the unit test file (.spec) for now.

Let's change the main application to include the newly added Angular component MyWebComponent. For this, we need to come back to the src/app directory.

```
cd angular-web-component/src/app
```

File: app.component.html (delete everything, just keep the one line)

```
<app-my-web-component></app-my-web-component>
```